

Database System Engineering and Distributed Backend Development

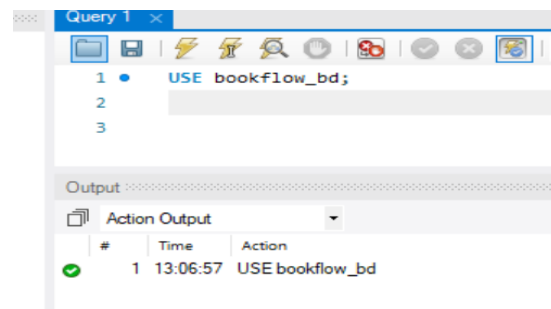
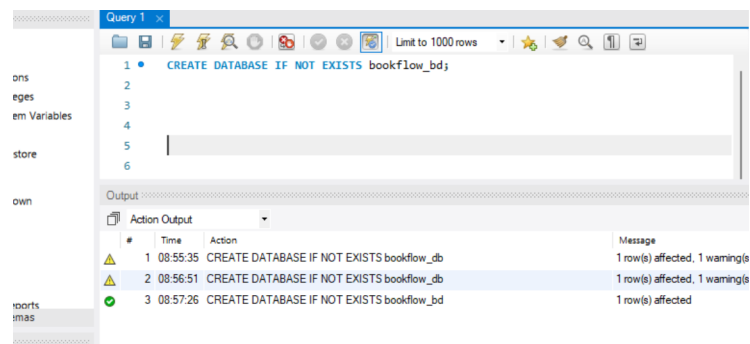
Course Code: 25CS1302E

WEEK2

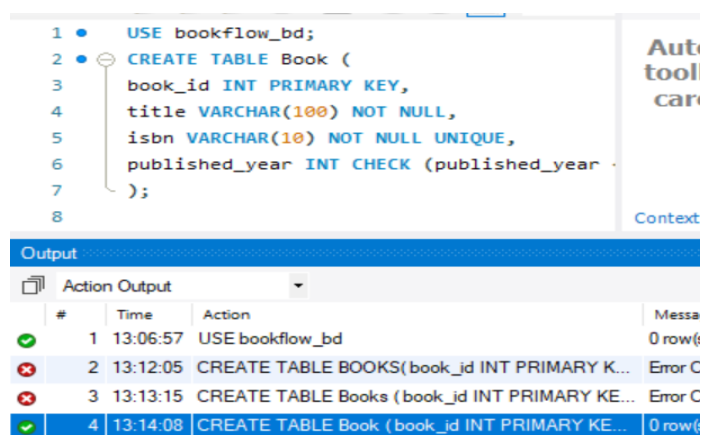
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K. Vaishnavi

Step1:Create database



Step2:Create Book table



Step3&4:Insert Books & Display

8

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```
INSERT INTO Book (book_id, title, isbn, published_year) VALUES
(1,'The Girl', '1234578909', 2025),
(2,'The Play Boy', '8907547893', 2008),
(3,'The Must', '1357218889', 2018);
```

12

Output

Action Output

#	Time	Action
✓ 1	13:06:57	USE bookflow_bd
✗ 2	13:12:05	CREATE TABLE BOOKS(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NU
✗ 3	13:13:15	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NU
✓ 4	13:14:08	CREATE TABLE Book (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NUL
✗ 5	13:22:39	INSERT INTO Book (book_id, title, isbn, published_year) VALUES ('The Girl', '12345
✓ 6	13:23:52	INSERT INTO Book (book_id, title, isbn, published_year) VALUES (1,'The Girl', '1234

DISPLAY:

8

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```
INSERT INTO Book (book_id, title, isbn, published_year) VALUES
(1,'The Girl', '1234578909', 2025),
(2,'The Play Boy', '8907547893', 2008),
(3,'The Must', '1357218889', 2018);
```

12

•

```
SELECT * FROM Book;
```

13

Result Grid

Filter Rows:

Edit:

Export/Import:

	book_id	title	isbn	published_year
▶	1	The Girl	1234578909	2025
	2	The Play Boy	8907547893	2008
	3	The Must	1357218889	2018
*	NULL	NULL	NULL	NULL

Output

Action Output

#	Time	Action	Message
✓ 1	13:06:57	USE bookflow_bd	0 row(s)
✗ 2	13:12:05	CREATE TABLE BOOKS(book_id INT PRIMARY KEY, title VARCHAR(10...	Error C
✗ 3	13:13:15	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100...	Error C
✓ 4	13:14:08	CREATE TABLE Book (book_id INT PRIMARY KEY, title VARCHAR(100)...	0 row(s)
✗ 5	13:22:39	INSERT INTO Book (book_id, title, isbn, published_year) VALUES ('The G...	Error C
✓ 6	13:23:52	INSERT INTO Book (book_id, title, isbn, published_year) VALUES (1,'The ...	3 row(s)
✓ 7	13:25:10	SELECT * FROM Book LIMIT 0, 1000	3 row(s)

Step5&6 & 7: Members Table - Create , Insert , Display

```
14 ● ○ CREATE TABLE Member (  
15     member_id INT PRIMARY KEY,  
16     full_name VARCHAR(100) NOT NULL,  
17     email VARCHAR(100) UNIQUE  
18 );  
19 ● INSERT INTO Member (member_id, full_name, email) VALUES  
20     (101, 'Vaishnavi', 'Vaishnavi@exam.com'),  
21     (102, 'Kushal', 'Kushal@exam.com'),  
22     (103, 'Sai', 'Sai@exam.com');  
23 ● SELECT * FROM Member;
```

Result Grid | | Filter Rows: | Edit:

	member_id	full_name	email
▶	101	Vaishnavi	Vaishnavi@exam.com
	102	Kushal	Kushal@exam.com
	103	Sai	Sai@exam.com
*	NULL	NULL	NULL

Member 11 ×

Output

Action Output ▼

	#	Time	Action
✓	10	13:33:12	CREATE TABLE Member (member_id INT PRIMARY KEY,
✓	11	13:35:20	INSERT INTO Member (member_id, full_name, email) VALU
✓	12	13:35:38	SELECT * FROM Member LIMIT 0, 1000

Step 8: Create Loan Table

```

24 ● CREATE TABLE Loans (
25     loan_id INT PRIMARY KEY,
26     member_id INT,
27     book_id INT,
28     loan_date DATE,
29     FOREIGN KEY (member_id) REFERENCES Member(member_id),
30     FOREIGN KEY (book_id) REFERENCES Book(book_id)
31 );
32

```

Context

Output

Action Output

#	Time	Action	Message
✓ 11	13:35:20	INSERT INTO Member (member_id, full_name, e...	3 row(s) affected Records: 3
✓ 12	13:35:38	SELECT * FROM Member LIMIT 0, 1000	3 row(s) returned
✗ 13	18:53:09	CREATE TABLE Loans (loan_id INT PRIMARY ...	Error Code: 1064. You have
✓ 14	18:54:34	CREATE TABLE Loans (loan_id INT PRIMARY ...	0 row(s) affected

Step9 & 10: Insert 10 loans & Display

```

32 ● INSERT INTO Loans (loan_id, member_id, book_id, loan_date)
33     (1,101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
34     (3,103, 3, '2025-01-10'), (4, 101, 2, '2025-02-09'),
35     (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-03'),
36     (7, 101, 3, '2025-09-08'), (8, 102, 3, '2025-03-08'),
37     (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-03-09');
38

```

Context

Output

Action Output

#	Time	Action	Message
✓ 1	19:27:52	INSERT INTO Loans (loan_id, member_id, book_id, l...	10 row(s) affected Records: 10

Display:

39 • `SELECT * FROM Loans;`

40

41

Result Grid						 Filter Rows:		Edit:		
	loan_id	member_id	book_id	loan_date						
▶	1	101	1	2025-01-05						
	2	102	2	2025-01-08						
	3	103	3	2025-01-10						
	4	101	2	2025-02-09						
	5	102	1	2025-02-05						
	6	103	2	2025-02-03						
	7	101	3	2025-09-08						
	8	102	3	2025-03-08						
	9	103	1	2025-03-15						
	10	101	1	2025-03-09						
Loans 12 ×										
Output										
 Action Output 										
	#	Time	Action							
✓	1	19:27:52	INSERT INTO Loans (loan_id, member_id, book_id,...							
✓	2	19:30:08	SELECT * FROM Loans LIMIT 0, 1000							

Step 11: — JOIN Query (Who Borrowed What)

41 • `SELECT m.full_name AS Member_name, b.title AS Book_Title`
 42 `FROM Loans l`
 43 `INNER JOIN Member m ON l.member_id = m.member_id`
 44 `INNER JOIN Book b ON l.book_id = b.book_id;`

Result Grid

Member_name	Book_Title
Vaishnavi	The Girl
Vaishnavi	The Play Boy
Vaishnavi	The Must
Vaishnavi	The Girl
Kushal	The Play Boy
Kushal	The Girl
Kushal	The Must
Sai	The Must
Sai	The Play Boy
Sai	The Girl

Result 13 x Read Only Context I

Output

Action Output

#	Time	Action	Message
3	19:36:21	SELECT m.full_name AS Member_name, b.title A...	Error Code: 1146. Table 'bookflow_br
4	19:38:17	SELECT m.full_name AS Member_name, b.title A...	10 row(s) returned

Step12 : GROUP BY Query

46 • `SELECT published_year, COUNT(book_id) AS Total_Books`
 47 `FROM Books`
 48 `GROUP BY published_year`
 49 `ORDER BY published_year;`
 50

Result Grid

published_year	Total_Books
1988	1
2008	1
2018	1

Output

Action Output

#	Time	Action	Message
1	19:27:52	INSERT INTO Loans (loan_id, member_id, book_id,...	10 row(s) affected F
2	19:30:08	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
3	19:36:21	SELECT m.full_name AS Member_name, b.title AS ...	Error Code: 1146. T
4	19:38:17	SELECT m.full_name AS Member_name, b.title AS ...	10 row(s) returned
5	19:44:42	SELECT published_year, COUNT(book_id) AS Tota...	3 row(s) returned

Step13 - 1 6:- Donation via TRANSACTION

```

51 • CREATE TABLE Donation_History(
52     donation_id INT PRIMARY KEY,
53     book_id INT,
54     donor_name VARCHAR(100),
55     donation_date DATE,
56     FOREIGN KEY (book_id) REFERENCES Book(book_id)
57 );
58

```

Output

Action Output

#	Time	Action	Message
3	19:36:21	SELECT m.full_name AS Member_name, b.title A...	Error Code: 1146.
4	19:38:17	SELECT m.full_name AS Member_name, b.title A...	10 row(s) returned
5	19:44:42	SELECT published_year, COUNT(book_id) AS To...	3 row(s) returned
6	21:36:32	CREATE TABLE Donation_History(donation_id l...	0 row(s) affected

```

58 • START TRANSACTION;
59
60 • INSERT INTO Book (book_id, title, isbn, published_year) VALUES
61     (4, 'Animal', '9875432178', 2008);
62
63 • INSERT INTO Donation_History(donation_id, book_id, donor_name, donation_date) VALUES
64     (1, 4, 'Vaishu', CURDATE());
65
66 • COMMIT;
67

```

Output

Action Output

#	Time	Action	Message
8	21:43:39	START TRANSACTION	0 row(s) affected
9	21:43:42	INSERT INTO Book (book_id, title, isbn, published_year) VALUES (4, 'Animal', '9875432178', ...	1 row(s) affected
10	21:43:45	INSERT INTO Donation_History(donation_id, book_id, donor_name, donation_date) VALUES ...	1 row(s) affected
11	21:43:47	COMMIT	0 row(s) affected

```

66 • COMMIT;
67 • SELECT * FROM Donation_History;
68
69

```

Result Grid | Filter Rows: | Edit:

donation_id	book_id	donor_name	donation_date
1	4	Vaishu	2026-08-09
NULL	NULL	NULL	NULL

Step 17 & 18: Index on ISBN + Fast Search

69 • CREATE INDEX idx_books_isbn ON Book(isbn);

70

71 • SELECT * FROM Book WHERE isbn = '9875432178';

Result Grid

Filter Rows:

Edit:

Export/Import:

	book_id	title	isbn	published_year
▶	4	Animal	9875432178	2008
*	NULL	NULL	NULL	NULL

Book 17 ×

Output

Action Output

#	Time	Action	Message
✓ 13	21:52:07	SELECT * FROM Book WHERE isbn = '9875432178' LIMIT ...	1 row(s) returned
✓ 14	21:52:25	CREATE INDEX idx_books_isbn ON Book(isbn)	0 row(s) affected R
✓ 15	21:52:29	SELECT * FROM Book WHERE isbn = '9875432178' LIMIT ...	1 row(s) returned

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